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Specific rate constants $k(E, J)$ and product state distributions in simple bond fission reactions. II. Application to $\text{HOOH} \rightarrow \text{OH} + \text{OH}$

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Detailed and simplified statistical adiabatic channel calculations of specific rate constants $k(E, J)$ and product quantum state distributions for the simple bond fission reaction $\text{HOOH} \rightarrow 2 \text{OH}$ are compared with recent measurements of state-resolved dissociation rates, product state distributions, and thermally averaged rate coefficients. A simple modification of phase space theory based on the statistical adiabatic channel model successfully predicts product state distributions and rate constants as well. Because of the amount of experimental data and theoretical analysis available, the dissociation of hydrogen peroxide is becoming a model case for simple unimolecular bond fission processes.

I. INTRODUCTION

Experimental studies of partially state-selected unimolecular reactions of neutral molecules have advanced considerably in recent years^{1–12} as have detailed theoretical models that relate state-resolved observables to the properties of a potential energy surface.^{13–23} Despite these achievements, neither fully state-resolved measurements of the rates of product formation for a particular total energy E and total angular momentum J nor rigorous quantum dynamical calculations on realistic and complete potential energy surfaces are generally possible. Under favorable conditions, measurements provide energy-resolved specific rate constants $k(E)$ along with some information about the rotational states J involved and the distribution of the products among their quantum states. Averaging resolved quantities over a thermal distribution of states permits comparison with high pressure dissociation or recombination rate constants and rate constants for other thermal reactions that proceed by complex formation. Classical trajectory calculations are one theoretical approach to unimolecular reactions that can employ realistic potential energy surfaces.^{21–23} A variety of statistical theories, such as variational transition state theory^{19,20} or statistical adiabatic channel models,^{13–18} provide alternative descriptions of such processes. A relation between the kinetic parameters and specific properties of the potential energy surface is a desideratum of all of these investigations.¹⁸

Detailed experimental and theoretical studies of as many aspects as possible of a few model reactions are potentially very useful. The first example of such a model system was the simple bond fission $\text{NO}_2 \rightarrow \text{NO} + \text{O}$ in the ground electronic state $\text{NO}_2(X^2A)$. Thermally averaged rate constants are available for high pressure dissociation and recombination, for isotope exchange, and for energy transfer

$\text{O} + \text{NO}(v', j') \rightarrow \text{O} + \text{NO}(v'', j'')$.²⁴ Population of specific states above the dissociation energy of the electronic ground state is possible in this molecule by excitation of electronic states that are mixed with the ground state. Although no time-resolved measurements of the specific rate constants $k(E)$ have been made for this system, they have been derived from Stern–Volmer analysis of collisional quenching in the photolysis. Product state distributions from early photofragmentation experiments seem to agree with the predictions of the statistical adiabatic channel model,¹³ but new measurements²⁵ are apparently in conflict with this analysis, either because of short dissociation lifetimes or complications from incomplete mixing of the electronic states.

Vibrational overtone excitation is an alternative activation mechanism that avoids electronic state mixing problems. Recent experiments on the simple dissociation $\text{HOOH} \rightarrow 2 \text{OH}$ in the electronic ground state^{7–10,26} using this technique provide data for comparison with a model study of this simple bond fission in a small molecule on the ground potential energy surface. Both detailed product state distributions^{7–9} and measurements of $k(E)$ by time-resolved product detection²⁶ and by linewidth analysis¹⁰ are available for the reaction initiated by vibrational overtone excitation as are trajectory calculations on simplified potential surfaces.^{22,23} The molecule is small enough to make detailed *ab initio* calculations²⁷ of the potential feasible as well. Both variational transition state theory^{19,20} and the statistical adiabatic channel model^{8,9} have been applied to the decomposition of HOOH using simplified potentials, the latter to predict the observed product state distributions. Here we describe detailed¹³ and simplified^{15–17} statistical adiabatic channel calculations of specific rate constants $k(E, J)$ and product state distributions and compare them to the measurements. The simplified statistical adiabatic channel model described here is a generalized and expanded version of that given in part I of this series.¹⁶ We use very simple models of the potential^{13–17} in the present study but have as our eventual goal the use of simple switching potentials for the long and short range behavior¹⁸ and, ultimately, *ab initio*

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potentials.²⁷ With the variety of experimental theoretical information that is becoming available, the simple bond fission $\text{HOOH} \rightarrow 2 \text{OH}$ is potentially a prototypical unimolecular dissociation reaction.

II. DETAILED STATISTICAL ADIABATIC CHANNEL MODEL

A. Calculation procedure

Our detailed calculations use the statistical adiabatic channel model (SACM) in its original version¹³ with the channel energies represented by a simple exponential interpolation scheme. Adiabatic channel potential curves $V_a(q)$ come from correlating reactant $[E_a(q_e, n)]$ and product $[E_a(\infty, p)]$ eigenstates using the approximate expression for the channel energies,

$$E_a(q) = E_a(q_e, n) \exp[-\alpha(q - q_e)] + E_a(\infty, p) \times \{1 - \exp[-\alpha(q - q_e)]\} + E_{\text{cent}}(q), \quad (2.1)$$

and by approximating the channel potential

$$V_a(q) = V(q) + E_a(q) \quad (2.2)$$

using a Morse potential

$$V(q) = D\{1 - \exp[-\beta(q - q_e)]\}^2 \quad (2.3)$$

and taking the centrifugal energy to be

$$E_{\text{cent}}(q) = B_{\text{cent}}(q)P(P+1), \quad (2.4)$$

where a rotation pseudoquantum number P is interpolated as

$$P = J \exp[-\alpha(q - q_e)] + L\{1 - \exp[-\alpha(q - q_e)]\}. \quad (2.5)$$

The notation is identical to that in the original formulation.^{13,15-17} The general statistical expression for the rate constant is

$$k(E, J) = W(E, J)/h\rho(E, J), \quad (2.6)$$

where $W(E, J)$ is the number of adiabatic reaction channels for which the channel threshold energy E_{0i} , given by the maximum of $V_a(q)$, is smaller than E and where $\rho(E, J)$ is the rovibrational density of states of the molecule.

Hydrogen peroxide is nearly a symmetric top molecule,^{28(a)} and we use the effective rotational constant $B_e = (B_0 + C_0)/2$ in the centrifugal energy. In the quasidiatomic approximation, this is given by

$$B_{\text{cent}}(z) = B_e/(1 + a_1 z + a_2 z^2) \quad (2.7)$$

with $z = \beta(q - q_e)$, $a_1 = 2/\beta q_e$, and $a_2 = 1/(\beta q_e)^2$. The “ K rotor”^{28(b)} together with the hindered internal rotation and two oscillators²⁹ of HOOH (1387 and 1266 cm^{-1}) are the “disappearing oscillators” that correlate with the four product rotors of the two OH fragments ($B_\infty = 18.871 \text{ cm}^{-1}$). The 875 cm^{-1} oscillator of HOOH , which best corresponds to the O–O stretching mode, is the reaction coordinate, and, the HOOH oscillators at 3611 and 3599 cm^{-1} are individually correlated with the two OH stretching oscillators at 3735 cm^{-1} . We use the symmetry number $\sigma_e = 2$ for the approximate C_2 symmetry of the reactant. For the adiabatic channels, the average symmetry number is σ_∞

$= \sigma_1 \sigma_2 S$ with $\sigma_1 = \sigma_2 = 1$ and $S_\infty = 2$, except for the lowest channel state, for which we take $\sigma_\infty = 1$. This has the consequence that the number of open channels obtained neglecting symmetry is divided by two except for the lowest channel.

We determine the eigenvalues of the hindered internal rotor for a HOOH molecule with the fourth vibrational overtone ($5\nu_{\text{OH}}$) excited by numerically solving the corresponding torsional differential equation.³⁰ We use a threshold energy for dissociation of

$$E_0 = \Delta H_0^\circ \hat{=} 207.5 \text{ kJ mol}^{-1} = 17\,346 \text{ cm}^{-1},^{32}$$

which corresponds to a well depth along the reaction coordinate of $D = 19\,147 \text{ cm}^{-1}$, and calculate a Morse parameter of $\beta = 2.45 \text{ \AA}^{-1}$ from the O–O stretching force constant of 4.58 $\text{mdyn/\AA}$.²² We obtained a best fit value of $\alpha = 1.08 \text{ \AA}^{-1}$ that is close to the “standard value”¹⁷ of $\alpha \approx 1 \text{ \AA}^{-1}$ by varying it to obtain agreement with the thermal recombination data, as described below. This best fit value is also close to the “standard ratio”¹⁷ $\alpha/\beta = 0.50$. The calculation correlates the states of the four “disappearing oscillators” (the K rotor, torsion, and two vibrations) with OH rotational states in order of increasing energy without allowing the channel potential curves to cross. The asymptotic energies of the $\text{OH}(^2\Pi_{3/2})$ products are the values for each level including the coupling of the rotational and electronic orbital angular momentum.³³ As usual, the available states are restricted to obey the angular momentum coupling relations $|N_1 - N_2| \leq N_r \leq N_1 + N_2$ and $|N_r - L| \leq J \leq N_r + L$ where L is the orbital angular momentum quantum number for the separating fragments and N is the quantum number corresponding to the total angular momentum of a product molecule less its electron spin angular momentum. We assign N as $N = 1 + j$ in the calculation where j is the nuclear rotational angular momentum quantum number of a product molecule. Distributions $P(N, E, J)$ of the products of the dissociation of HOOH among their various quantum states N come from

$$P(N, E, J) = W(N, E, J)/W(E, J), \quad (2.8)$$

where $W(N, E, J)$ is the number of open channels correlating with products in the state N . This is exactly the approach previously used in deriving product state distributions for comparison to measurements.^{8,9}

There is an uncertainty with respect to the electronic states in the calculation. Two $\text{OH}(^2\Pi_{3/2})$ radicals can combine to form several electronic states of HOOH , of which one 1A_g state is strongly bound and the other 1A_u , 1B_g , and 1A_g states are weakly bound ($\leq 600 \text{ cm}^{-1}$).²⁷ Similarly, there are excited electronic states of HOOH correlating with OH in the $^2\Pi_{1/2}$ state, which is 139.7 cm^{-1} higher in energy than the $^2\Pi_{3/2}$ state.³² States from these excited electronic potentials could contribute to the reaction if electronically nonadiabatic behavior were to occur. We neglect such contributions although one cannot exclude collision-induced transitions between the various states in high pressure rate measurements. Fitting α to the recombination data partially compensates for any such effects.

TABLE I. Densities of states $\rho(E, J)$, sums of open channels $W(E, J)$, and rate coefficients $k(E, J)$ calculated using the detailed statistical adiabatic channel model with $\alpha/\beta = 0.44$.

$E - E_0$ (cm ⁻¹)	$\rho(E, J)$ (cm)	$W(E, J)$	$k(E, J)$ (s ⁻¹)
<i>J</i> = 0			
0	6.8	1	4.4×10^9
200	7.2	4.5	1.9×10^{10}
400	7.2	10	4.2×10^{10}
800	7.9	34.5	1.3×10^{11}
1200	8.9	60	2.0×10^{11}
1600	9.7	99	3.1×10^{11}
2000	11.0	134.5	3.7×10^{11}
2400	11.4	176	4.6×10^{11}
2800	12.2	225	5.5×10^{11}
<i>J</i> = 10			
0	131	0	0
200	139	19.5	4.2×10^9
400	143	88	1.8×10^{10}
800	159	319.5	6.0×10^{10}
1200	172	692	1.2×10^{11}
1600	186	1150.5	1.8×10^{11}
2000	204	1846	2.7×10^{11}
2400	220	2655	3.6×10^{11}
2800	240	3482.5	4.3×10^{11}
<i>J</i> = 40			
0	177	0	0
200	197	3	4.6×10^8
400	201	30	4.5×10^9
800	224	187.5	2.5×10^{10}
1200	251	499	6.0×10^{10}
1600	278	958	1.0×10^{11}
2000	303	1567	1.5×10^{11}
2400	334	2387	2.1×10^{11}
2800	367	3378	2.7×10^{11}

B. Results

Table I collects representative values of $\rho(E, J)$, $W(E, J)$, and $k(E, J)$ obtained for $\alpha = 1.08 \text{ \AA}^{-1}$ and $\beta = 2.45 \text{ \AA}^{-1}$ ($\alpha/\beta = 0.44$), and Fig. 1 shows the dependence of the rate constants on energy for a number of angular momenta. In the energy range shown, only rotational product states of OH are accessible. Figure 1 shows that $k(E, J)$ increases sharply above the threshold energies, as is typical for "loose" activated complexes. There is also a very pronounced influence of J on $k(E, J)$. One effect is that the centrifugal barrier increases the threshold energy as J increases. Also, the degeneracy arising from the K states of the symmetric top makes the threshold rate constants depend strongly on J . Because only a single channel is open at the reaction threshold, the sum of channels is $W(E, J) = 1$, and the rate constant [Eq. (2.6)] is determined solely by the density of states $\rho(E, J)$. In the high energy region well above threshold, the compensating contributions of the K states to $\rho(E, J)$ and $W(E, J)$ change the variation of the rate constant with J away from the monotonic behavior observed near the threshold energy, and the $k(E, J)$ curves cross. The sharp increase of the rate constant at threshold and strong dependence on J appears in earlier SACM calculations^{13,16} as well as in microcanonical variational transition state calculations.²⁰ Comparison with partially resolved or unresolved measurements requires averaging the detailed $k(E, J)$ over the distribution of E and J that is appropriate for a particular measurement. For example, averaging over a complete thermal distribution gives a high pressure recombination rate

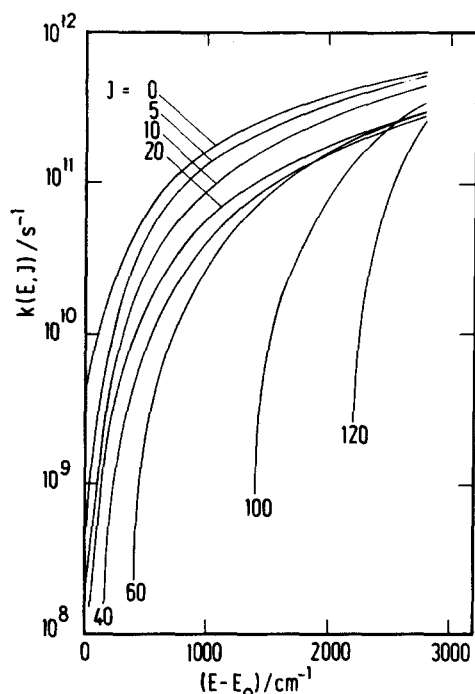


FIG. 1. Specific rate constants $k(E, J)$ for HOOH dissociation calculated using the detailed statistical adiabatic channel model with $\alpha = 1.08 \text{ \AA}^{-1}$ and $\beta = 2.45 \text{ \AA}^{-1}$ ($\alpha/\beta = 0.44$) for various angular momentum quantum numbers J . The total energy is E and the threshold energy is E_0 .

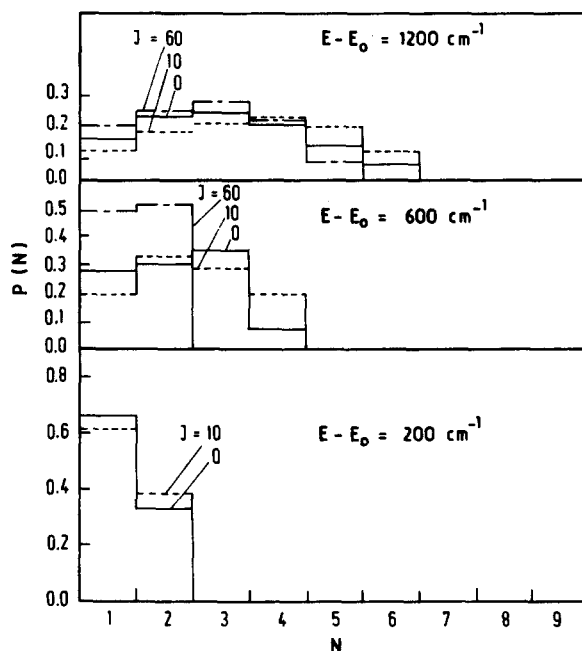


FIG. 2. The product state distributions $P(N)$ of OH from the dissociation of HOOH calculated using the detailed statistical adiabatic channel model with $\alpha/\beta = 0.44$ for different amounts of excess energy and total angular momentum. N is the quantum number corresponding to total OH angular momentum less electron spin angular momentum. In the calculation it is assigned as $N = 1 + j$ where j is the nuclear rotational angular momentum quantum number.

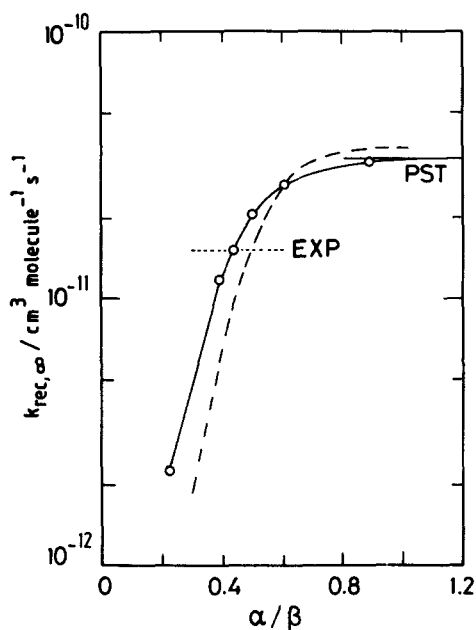


FIG. 3. High pressure recombination rate constants $k_{\text{rec},\infty}$ at 300 K for the reaction $2 \text{ OH} \rightarrow \text{HOOH}$ calculated using the detailed (solid line) and simplified (broken line) statistical adiabatic channel model as a function of the SACM parameter α/β . The phase theory result (PST) and the experimental measurement (EXP) (Ref. 38) are shown also.

constant of $k_{\text{rec},\infty}(T) = 1.5 \times 10^{-11} (T/300 \text{ K})^{-0.37} \text{ cm}^3 \text{ molecule}^{-1} \text{ s}^{-1}$ over the temperature range from 200 to 1500 K for $\alpha/\beta = 0.44$. Figure 2 shows the calculated distributions of the OH products among their rotational states. Close to the threshold energy there is a noticeable depen-

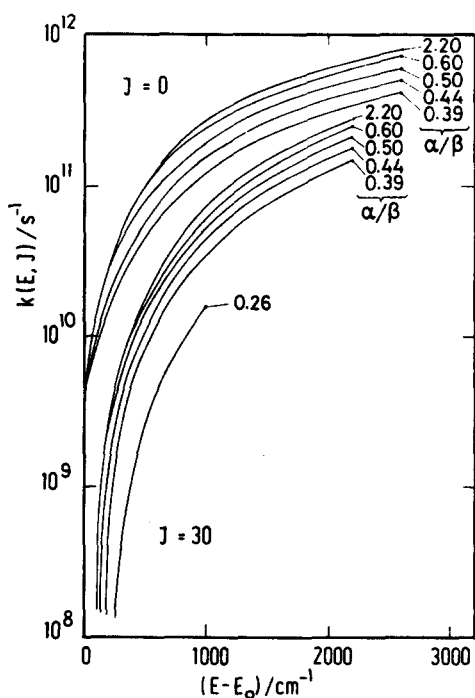


FIG. 4. Specific rate constants $k(E, J)$ for HOOH dissociation calculated using the detailed statistical adiabatic channel model for $J = 0$ and $J = 30$ for several different values of the SACM parameter α/β .

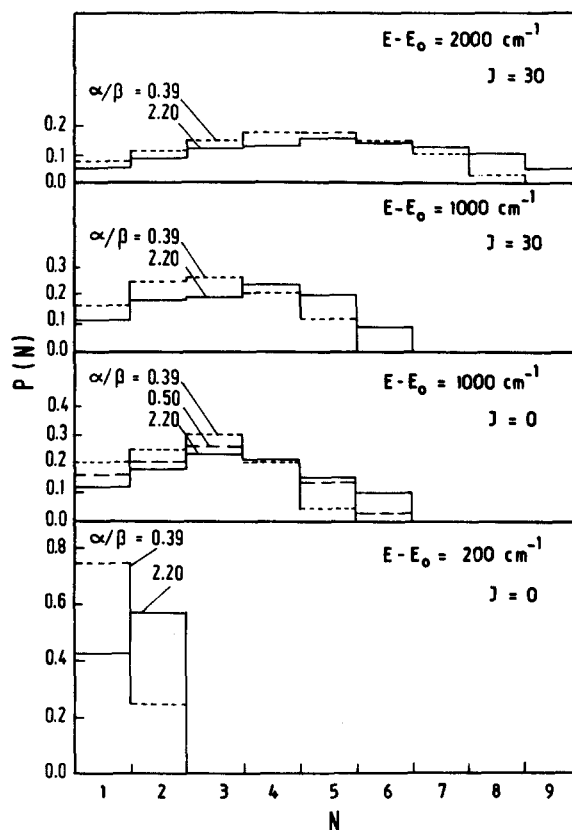


FIG. 5. The product state distributions $P(N)$ of OH from the dissociation of HOOH calculated using the detailed statistical adiabatic channel model for $J = 0$ and $J = 30$ for several different values of the SACM parameter α/β and excess energy.

dence of the distributions on the total angular momentum of HOOH, but with increasing energy the J dependence becomes much weaker. Small excess energies produce maxima at low rotational excitation of OH, and greater energy favors more product excitation.

The adjustable parameter α can be fit to one experimental observation and used in the SACM to predict other quantities. Figures 3–5 show the sensitivity of the recombination rate constant, of the detailed rate constant, and of the product state distributions to the value of α . For α/β exceeding unity, all of these results approach their phase space limits. The recombination rate constant $k_{\text{rec},\infty}$ (Fig. 3) decays markedly with decreasing α/β for $\alpha/\beta \lesssim 0.7$ and is quite sensitive around the “standard ratio”¹⁷ of 0.5. This limits the precision with which one can fix α by fitting $k_{\text{rec},\infty}$, as described below. The detailed rate constant $k(E, J)$ (Fig. 4) depends on α/β for $\alpha/\beta \lesssim 0.7$, as well, but above this value, it approaches the phase space theory result. The dependence of the product energy distributions (Fig. 5) on α/β is pronounced at energies close to the dissociation threshold but is considerably weaker at higher energies. In all cases smaller α/β ratios, corresponding to “more rigid activated complexes”, produce less energetic distributions. These properties of $k(E, J)$, $k_{\text{rec},\infty}$, and $P(N, E, J)$ are analogous to those observed in earlier detailed statistical adiabatic channel calculations.^{13–17,34}

III. SIMPLIFIED STATISTICAL ADIABATIC CHANNEL MODELS

A. General procedure

Applying the detailed SACM to a large molecule can present a major problem in counting all possible channels; however, a simplified version of the SACM,^{15,16} where the various coordinates are treated individually and average coupling corrections are applied *a posteriori*, is more tractable. Although one may formulate (or "derive") such global coupling corrections in a plausible way, the quality of the corrections can only be judged by comparison with detailed SACM calculations, as has been done for the NO₂ and C₂H₆ systems.^{15,16} Application of the simplified model to HOOH is a particularly stringent test. The large rotational constants of the fragments make it necessary to slightly modify the coupling corrections and to add quantum corrections to the classical treatment of rotor-type coordinates. In the following, we introduce quantum corrections to the number of channel functions W of rotor-type coordinates and show that the simplified SACM agrees well with the detailed calculations.

Treating the individual disappearing oscillators separately gives the contribution W of each to the total number of open channels $W(E, J)$. These functions are approximated by the numerical relationship¹⁵

$$W \approx [(E - D)/\epsilon]^x. \quad (3.1)$$

The two parameters are the effective quantum of the disappearing oscillator ϵ , which lies between that of the oscillator at equilibrium ϵ_e and the rotational constant of the fragment B_∞ , and the effective exponent x , which lies between the value for a harmonic oscillator $x = 1$ and that for a free rotor $x = 0.5$. The exponent depends strongly on α/β and weakly on the rotational constants of the molecule and fragments (B_e and B_∞) and on ϵ_e . Empirical relations in which a single average value of ϵ_e and B_∞ is used for all the disappearing oscillators¹⁵ give the parameters ϵ and x as a function of α/β . Comparison to the exact calculations shows that using the individual ϵ_e and B_∞ for each channel is actually a better approximation than using an average. In the previous development of the simplified SACM,¹⁶ the number of open channels for l disappearing oscillators with effective quanta ϵ_j and r free internal rotors with rotational constants B_i , which comes from a classical convolution of the rotors with the disappearing oscillators, is

$$W^l(E^*) = \frac{[\Gamma(1/2)]^r \prod_{j=1}^l \Gamma(1 + x_j)}{\Gamma(r/2 + 1 + \sum_{j=1}^l x_j)} \times \prod_{i=1}^r (E^*/B_i)^{1/2} \prod_{j=1}^l (E^*/\epsilon_j)^x. \quad (3.2)$$

This expression is the basis³⁵ for constructing the starting array of a Beyer-Swinehart counting algorithm³⁶ that is applied to the remaining "conserved" quantized harmonic oscillators to obtain the total number of open channels $W(E, J)$ uncorrected for conservation of angular momentum.

There are only two conserved oscillators in HOOH, and they have such large quanta that they contribute to the number of states only at energies that are relatively far above the

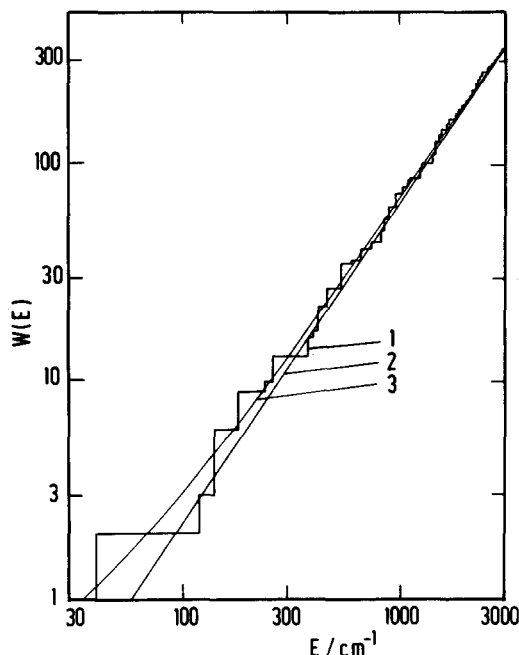


FIG. 6. Number of states $W(E)$ for the combination of a one-dimensional rotor ($B_1 = 20 \text{ cm}^{-1}$) and a two-dimensional rotor ($B_2 = 70 \text{ cm}^{-1}$). Curve (1) is directly counted number of states, curve (2) is a classical calculation, $W(E) = 2E^{2/3}/3B_2B_1^{1/2}$, and curve 3 is the calculation with the quantum expression, $W(E) = 2/3(E/B_2 + 1/2)(E/B_1 + 1/2)^{1/2}$.

reaction threshold energy, E_0 . However, the effective quanta ϵ_j of the disappearing oscillators are rather large, since they correlate with the large rotational quanta of the light OH fragments ($B_\infty = 18.871 \text{ cm}^{-1}$). Thus, the classical expression given above underestimates the number of open channels within a few hundred cm^{-1} of the threshold because it contains no contribution at threshold, where a real quantum oscillator has one state. Figure 6 illustrates this for a two-dimensional rotor convoluted with a one-dimensional rotor. The exact count of the states (curve 1) is above the classical approximation (curve 2) for excess energies up to several hundred cm^{-1} . The better approximation, shown as curve 3 in the figure, is given by a simple quantum expression that replaces $(E^*/B)^{1/2}$ by $(1/2 + E^*/B)^{1/2}$ and, thus, reduces the discrepancy with the exact count results. The quantum corrected version of Eq. (3.2) is

$$W^l(E^*) = \frac{[\Gamma(1/2)]^r \prod_{j=1}^l \Gamma(1 + x_j)}{\Gamma(r/2 + 1 + \sum_{j=1}^l x_j)} \times \prod_{i=1}^r (1/2 + E^*/B_i)^{1/2} \times \prod_{j=1}^l (1/2 + E^*/\epsilon_j)^x. \quad (3.3)$$

Adding $1/2$ to E^*/B_i and E^*/ϵ_j seems superior to the previously proposed correction of adding unity to the total classical sum of channels W .¹⁶ Equations (3.2) and (3.3) describe the convolution of a set of one-dimensional rotors neglecting angular momentum coupling constraints. The calculations shown in Fig. 6 include those constraints, which are briefly described below.

Two types of coupling corrections must be applied to the

$W(E, J)$ obtained from convoluting the number of channels for individual disappearing oscillators and rotors [Eq. (3.3)] with the conserved harmonic oscillators. One is an angular momentum factor $F_{AM}(E, J)$ that corrects for the couplings neglected in the factoring procedure that yields Eqs. (3.2) and (3.3).¹⁶ Omitting this factor amounts to neglecting the conservation of angular momentum, and it is the absence of this correction that makes conventional RRKM theory, without the inclusion of angular momentum conservation, inapplicable to simple bond fission reactions. The value of $F_{AM}(E, J)$ is interpolated between that for the reactant and that for the fragment $F_{AM, \infty}$ as described previously.¹⁶ In analogy with Eq. (3.3), a quantum correction is introduced in $F_{AM, \infty}$ by replacing $(E^*/B_\infty)^{1/2}$ with $(E^*/B_\infty + 1/2)^{1/2}$, and the number of conserved oscillators s_∞ excludes those whose quanta are larger than the total energy available.

The second correction comes from neglecting the excitation of other disappearing oscillators in the calculation for a particular oscillator. The energy in the other disappearing oscillators has the effect of modifying the underlying potential on top of which the channel eigenvalues for the particular disappearing oscillator are to be placed. In the original formulation, the correction for this effect took the form of replacing the actual β used in calculating the value of the effective oscillator quantum $\epsilon(\alpha/\beta)$ with an effective β given by

$$\beta_{\text{eff}}(E, J) = \beta [D/D_{\text{eff}}(E, J)]^{1/2}, \quad (3.4)$$

where D is the usual Morse dissociation energy along the reaction coordinate and D_{eff} is the corresponding parameter with an added average channel energy for all of the coordinates except the particular one being considered. Instead of the earlier simple expression for D_{eff} ,¹⁶ we now use a more general result that is presented in Appendix A.

The implementation of the simplified SACM for HOOH uses the conserved and disappearing oscillators that we identified for the detailed SACM calculation above except that the torsional oscillator, whose levels were found numerically in the detailed calculation, is replaced in the calculation of $W(E, J)$ by an oscillator having a frequency of 220 cm^{-1} . Because this approximation, which neglects the small *trans* torsional barrier in HOOH, produces a rovibrational density of states that is about 30% too high at energies close to or above the dissociation threshold, we use the direct count of exact hindered rotor states for $\rho(E, J)$, as in the exact calculation.

B. SACM-modified phase space theory (SACM-PST)

The simplified version of the SACM^{15,16} does not include a method for calculating the product energy distributions for comparison to highly resolved experiments. However, we can formulate a modified phase space theory that is consistent with the SACM and provides the requisite product state information. The analysis of a large number of experimental studies of simple bond fission systems by simplified SACM gives a fairly narrow range of experimental α/β values close to 0.5.¹⁷ On the other hand, the $x(\alpha/\beta)$ relation for W ¹⁵ gives $x \approx 0.5 \pm 0.05$ for $\alpha/\beta \geq 0.4$. Therefore, the

pattern of threshold energies $E_{0,i}$ for disappearing oscillators closely corresponds to free rotor states, although the effective rotational constants $\epsilon(\alpha/\beta)$ generally still differ markedly from the product rotational constants B_∞ . A SACM based modification of phase space theory (SACM-PST) using the effective rotational constant allows one to calculate specific rate constants and product state distributions.

The modification uses the usual number of states function $W(E, J)$ from phase space theory, which accurately accounts for angular momentum couplings at $q = \infty$, but modifies the rotational constants of the fragments according to the $\epsilon(\alpha/\beta)$ relationship.¹⁵ This gives $W(E, J)$ as approximately the smaller of the $(2J + 1)W(E^*, J = 0)$ and $W(E^*, \text{high } J)$ expressions, where the expressions for W are those described in the Appendix of part I.¹⁶ One can also apply quantum corrections by replacing $(E^*/B)^{1/2}$ with $(E^*/B + 1/2)^{1/2}$ or use accurate counting schemes for W . The simplest approach is to employ the $\epsilon(\alpha/\beta)$ relation and account for additional coordinate coupling with Eqs. (A1)–(A3) given in Appendix A. The SACM-PST is an adequate approximation in the absence of a good model of the potential energy. As in the detailed SACM calculations, applying the SACM-PST model to the HOOH system requires the correct asymptotic energies including the coupling of orbital and rotational angular momentum and spin-orbit coupling in the OH molecule.

C. Results

Having demonstrated the influence of the parameters α and β on the specific rate constants $k(E, J)$ (Fig. 4) and on the product distributions (Fig. 5), we now compare the results of a detailed SACM calculation with the simplified models. In the detailed calculation, we use the values $\alpha = 1.08 \text{ \AA}^{-1}$ and $\beta = 2.45 \text{ \AA}^{-1}$ ($\alpha/\beta = 0.44$) that reproduce the experimental $k_{\text{rec}, \infty}$ value of $1.5 \times 10^{-11} \text{ cm}^3 \text{ molecule}^{-1} \text{ s}^{-1}$ at 300 K. Comparison of the detailed and simplified calculations in Fig. 3 shows a discrepancy between the values of $k_{\text{rec}, \infty}$ obtained with the two models for the same value of α/β . However, adjusting the value of α/β from 0.44 to 0.50 brings the simplified SACM, as well as SACM-PST, into agreement with the detailed SACM. This makes the $W(E, J)$, and hence $k(E, J)$, as well as the $k_{\text{rec}, \infty}$ values the same for all three calculations. [A small difference remains in the calculated values of $k_{\text{rec}, \infty}$ in the phase space limit ($\alpha/\beta \gg 1$) because of the use of simplified state counting, classical rotational partition functions, and other approximations in the simplified model.¹⁶] Figure 7 compares $W(E, J)$ for the various treatments using $\alpha/\beta = 0.44$ for the detailed calculation and $\alpha/\beta = 0.5$ for the simplified calculations. Outside of the energy range shown in the figure, the use of the classical number of channel functions for the rotor-type coordinates decreases $W(E, J)$ at small excess energies. Using $\beta_{\text{eff}} = \beta$, on the other hand, increases $W(E, J)$ very slightly at moderate excess energies, whereas it decreases $W(E, J)$ at high excess energies. There is some compensation of errors as well as some unavoidable freedom in the procedure for determining β_{eff} . The SACM-PST modification of phase space theory performs as well for HOOH as the simplified SACM described above. Although the effect of using

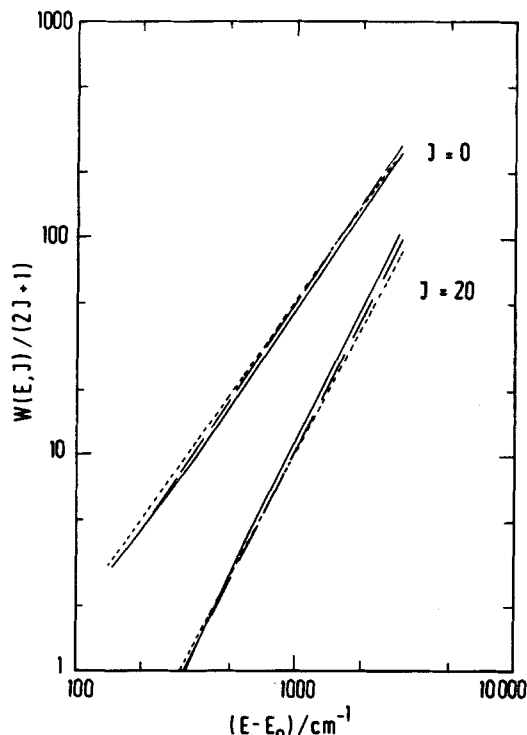


FIG. 7. Number of open channels for dissociation of $\text{HOOH} \rightarrow 2 \text{OH}$ calculated using the detailed SACM with $\alpha/\beta = 0.44$ (solid line), the simplified SACM with $\alpha/\beta = 0.5$ (dotted line), and the SACM modification of phase space theory (SACM-PST) with $\alpha/\beta = 0.50$ (dashed line). The modified rotational constants, in the simplified SACM calculation are $B_1 = 30.9 \text{ cm}^{-1}$ and $B_2 = 19.0 \text{ cm}^{-1}$.

$\beta_{\text{eff}} = \beta$ is not too large over the energy range shown, this correction becomes important at high energies, particularly for molecules with small dissociation energies.

Figure 8 compares OH product energy distributions from detailed SACM and SACM-PST calculations. The SACM-PST distributions come from a PST count using a rotational constant for OH that has been increased from 18.871 cm^{-1} to 24.2 cm^{-1} . This value is the geometrical mean of the quanta of the four disappearing oscillators which become OH rotors as given by the $\epsilon(\alpha/\beta)$ relationship with $\alpha/\beta = 0.5$. At small excess energies, the distributions predicted by the SACM and the SACM-PST agree exactly, but, at larger energies, the SACM-PST calculated for $\alpha/\beta = 0.44$ is slightly "cooler". Increasing α/β from 0.44 to 0.50, as we did above to bring the $k_{\text{rec},\infty}$ values from the detailed and the simplified SACM into agreement, produces a distribution that is nearly identical to the detailed results. *Both methods give nearly the same product distributions when α/β is fit from a measured $k_{\text{rec},\infty}$.*

This comparison of detailed and simplified SACM calculations confirms that the simplified versions reproduce the physical effects of the detailed SACM well and show only minor quantitative discrepancies. Fitting α/β to an experimental observable is an empirical solution to the difficulty of using an oversimplified potential energy surface in the calculation. Because the detailed SACM becomes complicated with increasing molecular complexity, the simplified SACM is a useful alternative in large systems. Further investigations have to show whether implementing more realistic po-

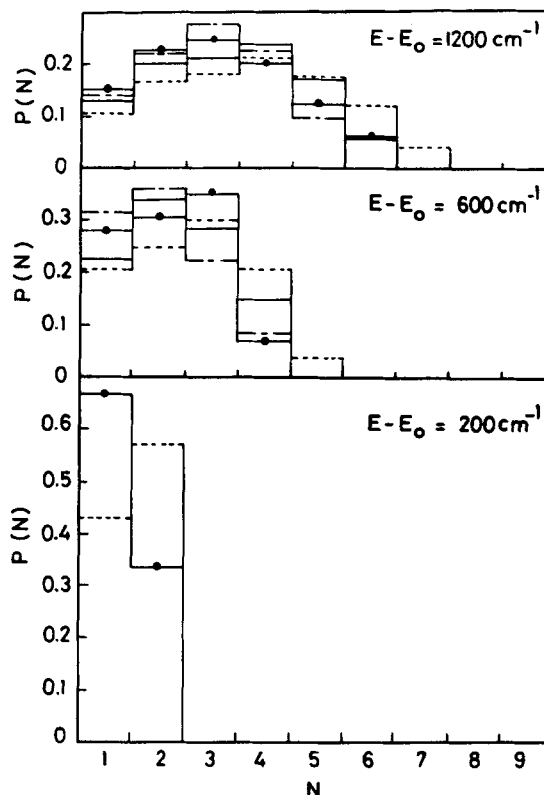


FIG. 8. The product state distributions $P(N)$ of OH from the dissociation of HOOH ($J = 0$) calculated using the detailed SACM, the SACM modification of phase space theory (SACM-PST), and phase space theory. The solid line with points is the detailed calculation with $\alpha/\beta = 0.44$, the solid line is the SACM-PST with $\alpha/\beta = 0.50$, the dot-dashed line is the SACM-PST with $\alpha/\beta = 0.44$, and the dashed line is phase space theory ($\alpha/\beta \geq 1$). For $\alpha/\beta = 0.50$, the average effective quantum is $\langle \epsilon \rangle = 24.2 \text{ cm}^{-1}$, and, for $\alpha/\beta = 0.44$, it is $\langle \epsilon \rangle = 30.0 \text{ cm}^{-1}$.

tential models^{18,27} will lead to changes that are larger than the difference between the detailed and simplified versions of the SACM.

IV. COMPARISON WITH EXPERIMENTS AND OTHER THEORETICAL MODELS

A. Comparison with experiments

The present treatment will lead to completely *ab initio* predictions of experimental observable only when accurate potential surfaces²⁷ are available. Using the "standard value"¹⁷ of $\alpha = 1.0 \pm 0.20 \text{ \AA}^{-1}$ for the looseness parameter or the standard value¹⁷ of the ratio $\alpha/\beta = 0.46 \pm 0.09$ for HOOH introduces a corresponding uncertainty of $\Delta \log k \approx \pm 0.2$ (Fig. 3) with analogous variations in $k(E, J)$ and $P(N, E, J)$ (Figs. 4 and 5). Empirical predictions cannot avoid this uncertainty, but fixing the α/β ratio by a fit to one experimental quantity, such as $k_{\text{rec},\infty}$ in the case of HOOH, allows the prediction of many details of the reaction, limited only by the precision of the experimental observation.

Fitting the high pressure recombination rate constant $k_{\text{rec},\infty} = 1.5 \times 10^{-11} \text{ cm}^3 \text{ molecule}^{-1} \text{ s}^{-1}$ for the room temperature thermal reaction³⁸



gives $\alpha/\beta = 0.44$ for the detailed SACM and $\alpha/\beta = 0.50$ for the simplified SACM. Because there is presently only a single measurement of the lower part of the fall-off curve for this reaction, extracting the rate constant requires a fairly long extrapolation toward the high pressure limit (see Appendix B). (More extensive measurements of k_{rec} , that will obviate this difficulty are in preparation in our laboratory.) The resulting factor of 2 uncertainty in $k_{\text{rec},\infty}$ produces a range of α/β from 0.35 to 0.71 (Fig. 3). A prediction of $k_{\text{rec},\infty}$ based on the standard value given above agrees with the measurement to better than the experimental uncertainty. In addition to the recombination experiments,³⁸ there are high pressure measurements for the HOOH thermal dissociation³⁹ that also reach into the transition to the high pressure regime. The simplified calculations using $\alpha/\beta = 0.5$ and the detailed calculations using $\alpha/\beta = 0.44$ are consistent with the value of k_{diss} , and, hence, with the room temperature value of $k_{\text{rec},\infty}$. (Appendix B gives the details of calculating the fall-off curves for dissociation and recombination over a wide range of temperatures.)

The thermal data determine $\alpha/\beta = 0.44$ in the exact SACM, and we can compare the $k(E,J)$ calculated using this value with recent measurements.^{10,26} The linewidth¹⁰ in the fifth vibrational overtone region ($6\nu_{\text{OH}}$) of jet cooled HOOH gives an upper limit to the specific rate constant $k(E)$ for the low values of J excited in the experiment, probably no greater than $J = 5$, that agrees with our calculated $k(E,J)$ (open triangle in Fig. 9). A comparison with the direct time resolved measurement²⁶ of the dissociation lifetime after excitation in the fourth ($5\nu_{\text{OH}}$) vibrational overtone region of room temperature HOOH is more difficult. Because excitation of the fourth overtone vibration adds insufficient energy for an initially unexcited HOOH molecule to react, only those that initially possess some thermal energy decompose. In addition, a number of different rotational states are excited by the vibrational overtone transition. If all of the molecules that initially possess enough thermal energy to react following excitation of the $5\nu_{\text{OH}}$ transition were excited, their average excess energy would be $\langle E^* \rangle \approx 415 \text{ cm}^{-1}$, and their average angular momentum would be $\langle J \rangle \approx 25$.⁸ (The average excess energy is considerably less than the 1600 cm^{-1} of energy characteristic of a full room temperature distribution, and the average angular momentum is somewhat larger than the corresponding room value of $15\hbar$.) These values only roughly describe the actual situation since the collection of states prepared depends on the details of the vibrational overtone spectroscopy. Nonetheless, using this crude estimate of the excess energy, we see that the fastest decay measured in the time resolved experiment gives a rate constant (closed triangle in Fig. 9) that is consistent with the SACM prediction for $J \approx 15$.

Thermal averaging over energies and angular momenta is present in the experiments to different extents. Canonical averaging for the thermal recombination experiments is well defined, and experiments on jet cooled molecules allow for selective excitation of very low J states. Identifying the initial rotational state in room temperature vibrational overtone initiated reactions requires the more difficult analysis of the rotational structure of the bands, and this problem presently

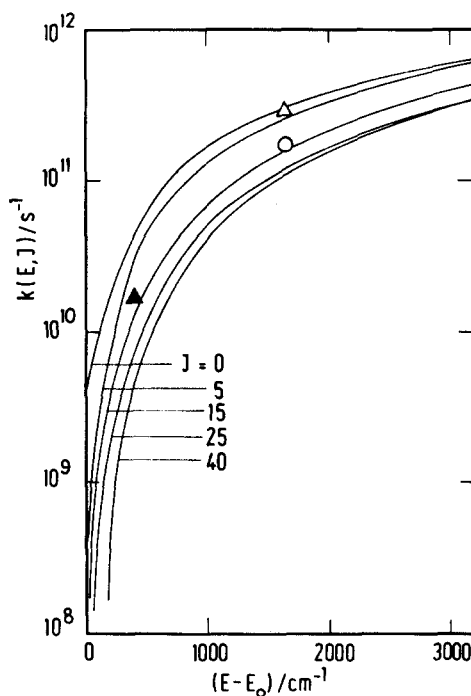


FIG. 9. Calculated and measured specific rate constants $k(E,J)$ for HOOH dissociation. The solid lines are the results of the detailed statistical adiabatic channel model with the fit value of $\alpha/\beta = 0.44$ derived from the high pressure thermal recombination rate constant $k_{\text{rec},\infty} = 1.5 \times 10^{-11} \text{ cm}^3 \text{ molecule}^{-1} \text{ s}^{-1}$ (Ref. 38). The open triangle is the upper limit inferred from the linewidth of the fifth vibrational overtone transition ($6\nu_{\text{OH}}$) in jet cooled HOOH (Ref. 10). The closed triangle is the time resolved measurement of product formation in a room temperature sample following excitation of the fourth vibrational overtone transition ($5\nu_{\text{OH}}$) (Ref. 26). The open circle is the result of a classical trajectory calculation (Ref. 22) for $J = 0$ at the energy of the fifth overtone vibration.

limits the rotational state specificity of room temperature measurements. Although uncertainties remain in the measurements of both the recombination rate constant and the detailed rate constants, the consistency of the different experiments and the calculation is quite satisfactory.

Comparison of the product state distributions predicted by the detailed SACM calculations with the measurements shows good agreement for decomposition initiated by excitation of either the fourth ($5\nu_{\text{OH}}$) or fifth ($6\nu_{\text{OH}}$) overtone vibration.^{8,9} The comparison is particularly revealing in the former case, where the reaction occurs very near the threshold energy, since PST predicts considerably more excitation than observed experimentally or predicted by the SACM. The differences between the two calculations are less at the higher energy excitation ($6\nu_{\text{OH}}$) where, as Fig. 5 illustrates, the distributions depend only weakly on α/β ^{37,41} and are less sensitive to the details of the calculation.

B. Comparison with other theoretical models

Hydrogen peroxide has recently been the subject of other statistical calculations^{19,20} and of classical trajectory calculations.^{22,23} Wardlaw and Marcus^{19,20} report a microcan-

onical variational transition state calculation that has a number of features in common with the SACM approach described here. The two types of calculations are roughly, but not completely, equivalent when applied on the same potential.¹³ The numbers of states $W(E, J)$ that they calculate using microcanonical transition state theory are roughly the same as obtained in the present work. One should, however, notice that their $W(E, J)$ contains the additional factor $2J + 1$ for spatial orientation of the total angular momentum. Also, it does not contain a symmetry number of a factor of 2. The factor $2J + 1$ cancels in $k(E, J)$, since $\rho(E, J)$ contains the same factor. In the present work, like in our earlier work, this factor always is omitted.

It is also interesting to compare the SACM calculations to the RRKM lifetime estimates of about 400 ps at threshold and of about 60 ps for 2000 cm^{-1} of excess energy that were made in conjunction with the time resolved measurements of the decomposition of $\text{HOOH}(5_{\text{OH}})$.²⁶ The calculated lifetime at 2000 cm^{-1} of excess energy is substantially longer than our SACM estimate of about 2.5 ps (Fig. 9). Because the two calculations are within a factor of 2 at threshold, apparently the RRKM calculation has used a rigid activated complex that produces a smaller number of states above threshold than given by our model with a relatively loose activated complex.

Classical trajectory calculations by Uzer *et al.*²² produce a calculated lifetime for $J = 0$ (open circle in Fig. 9) that lies within a factor of 2 of our statistical calculation. This is particularly interesting because they find incomplete energy redistribution in which most of the vibrations in the molecule are strongly coupled on the time scale of the decomposition *except* for the initially unexcited OH oscillator, which re-

mains unexcited during the reaction. The agreement with the statistical calculation appears to come from the high frequency oscillator being effectively excluded in the statistical calculation as well. In the statistical theory, the large quanta of the OH stretches contribute negligibly to the number of states at excess energies less than about 3500 cm^{-1} , and, hence, both calculations deal with the strongly coupled low frequency vibrations. Another classical trajectory calculation²³ performed at much higher excitation, $16\nu_{\text{OH}}$, gives reaction lifetimes that are consistent with our estimates, as the comparison with the simplified SACM results and the trajectory results in Fig. 10 shows. The most detailed comparisons will be between trajectory and statistical calculations that are carried out using the same potential surface. One such comparison, for a reaction controlled by an ion-dipole potential, shows that the two agree well for conditions under which the classical treatment is adequate.⁴²

V. CONCLUSIONS

Comparison of detailed statistical adiabatic model calculations with a variety of new experimental results shows good agreement with the detailed rate constants and the general features of the product quantum state distributions in two energy ranges. The calculations use a single potential parameter α/β that is fit to an *independent* measurement of the high pressure thermal recombination rate constant. These detailed SACM calculations also permit an extensive comparison and evaluation of more tractable simplified versions of the SACM. The simplified SACM, when appropriately modified to account for the large rotational constant of the products, agrees well with the detailed calculations. Fitting the simplified SACM to the same thermal rate constant data as used with the detailed SACM gives a different potential parameter ($\alpha/\beta = 0.50$ compared to 0.44 for the detailed model) but produces essentially identical rate constants. A modification to phase space theory based on the SACM (SACM-PST), which was introduced to obtain product state distributions conveniently, also agrees quite well with the detailed formulation. In fact, when the potential parameter is fit to thermal recombination data, the calculations are equally good for the detailed SACM, the simplified SACM, and the SACM modified phase space theory.

The participation of weakly bonding excited states of HOOH that correlate with the $^2\Pi_{3/2}$ and $^2\Pi_{1/2}$ states of OH is a remaining uncertainty in the calculation that is probably compensated in part by fitting the potential parameter to thermal data. A more rigorous approach also requires the use of realistic potential energy surfaces that include the long range dipole-dipole interaction between the OH fragments. Potential switching models,¹⁸ techniques for implementing arbitrary potentials, and *ab initio* potential energy surface calculations²⁷ promise to make calculations with no fitted parameters possible. The growing body of detailed experimental and theoretical information on the HOOH bond fission make it a particularly attractive model system on which to test theory against new experimental developments.

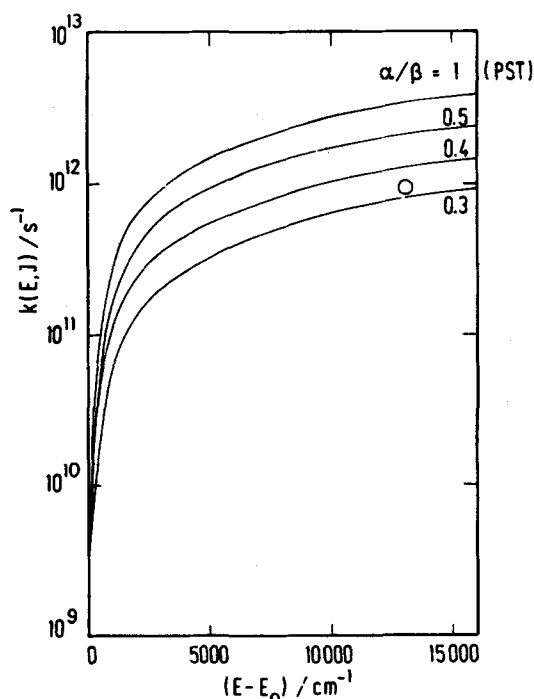


FIG. 10. Specific rate constants $k(E, J)$ for HOOH dissociation calculated using the simplified statistical adiabatic channel model with $\alpha/\beta = 0.50$. The open circle is a classical trajectory calculation (Ref. 23).

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APPENDIX A: EFFECTIVE MORSE POTENTIAL FOR DISAPPEARING OSCILLATORS

The threshold energy pattern of disappearing oscillators has been analyzed by superimposing the channel energy $E_a(q)$ of a single disappearing oscillator, Eq. (2.1), on the Morse potential $V(q)$. If other conserved or disappearing oscillators are excited, their channel energies $E'_a(q)$ are superimposed on the Morse potential as well. As a first approximation, we assume that the sum of $E'_a(q)$ and $V(q)$ is a Morse-type function and obtain the average excitation $E'_a(q)$ through equipartitioning of the energy in excess of the bond energy, $E - E_0$, among all conserved and disappearing oscillators near the equilibrium position q_e . Figure 11 shows the resulting situation, neglecting centrifugal effects. The lowest curve is the Morse potential $V(q)$, and the middle curve is the sum of $V(q)$ and the average channel energy $E'_a(q)$. The top curve is the energy of the highest open channel. The channel energies of conserved and disappearing oscillators change differently as $q \rightarrow \infty$. They are for oscillator to oscillator transformations, $E'_a(\infty) = E'_a(q_e) (\epsilon_\infty / \epsilon_e)$,

for a K rotor to free rotor,

$$E'_a(\infty) = E'_a(q_e) [B_\infty / (A_e - B_e)],$$

and, for oscillator to rotor transformations, $E'_a(\infty) = B_\infty [E'_a(q_e) / \epsilon_e]^2$. The last transformation comes from the adiabatic conservation of a channel quantum number n on the way from $q = q_e$, where $E'_a(q_e) = (n + 1/2)\epsilon_e \approx n\epsilon_e$, to $q \rightarrow \infty$ where $E'_a(\infty) = n(n + 1)B_\infty \approx n^2 B_\infty$. These relations connect $E_a(q_e) + E'_a(q_e)$ with the excess energy $E - E_0$. Taking the sum of the energies, $E_a(q_e) + E'_a(q_e)$, to be equilibrated among the oscillators yields a partitioning between the average energies $\langle E'_a(q_e) \rangle$ and $\langle E_a(q_e) \rangle$. The transformations described above give the corresponding value of $\langle E'_a(\infty) \rangle$. Thus, the effective well depth for the particular disappearing oscillator is $D + \langle E'_a(\infty) \rangle - \langle E'_a(q_e) \rangle$.

An additional modification of the potential by rotation of the molecule enters as the difference between the increase of the threshold energy for a particular angular momentum state J , $E_0(J) - E_0$, and the rotational energy $B_e J(J + 1)$. Thus we have an effective well depth of

$$\begin{aligned} D_{\text{eff}}(E, J) = & D + E_0(J) - E_0 - B_e J(J + 1) \\ & + [(b - 1)/b] (E - E_0) \\ & + [M/b - (s - 2)/(s - 1)] M/2N \\ & \times \{ [1 + (E - E_0)4N/M^2]^{1/2} - 1 \}, \end{aligned} \quad (\text{A1})$$

where

$$M = 1/(s - 1) \sum_i \epsilon_{\infty i} / \epsilon_{ei} \quad (\text{A2})$$

and

$$N = 1/(s - 1)^2 \sum_j B_{\infty j} / (\epsilon_{ej})^2. \quad (\text{A3})$$

The parameter s is the number of internal coordinates including the K rotor but excluding the reaction coordinate, and b is the number of disappearing oscillators excluding the K rotor. The sum over i includes all of the conserved oscillators and free rotors (including the K rotor), for which $E'_a(\infty)$ depends linearly on $E'_a(q_e)$, and the sum over j includes all $b - 1$ disappearing oscillators except the particular one being explicitly considered. For that oscillator, ϵ comes from the $\epsilon(\alpha/\beta)$ relationship. The expression for the effective well depth, Eq. (A1), improves on the previous expression [Eq. (2.13) in part I]¹⁶ because it includes the simultaneous change in the quanta of the disappearing and conserved oscillators along the reaction coordinate. At high energies, the previous expression¹⁶ may produce unphysical behavior. A further improvement on Eq. (A1) is to include the few special cases that occur, such as the omission of the K rotor for $J = 0$ and the omission of all coordinates that have quanta that are too large for excitation at the energy being considered. Although this treatment of the effective potential in obtaining the effective Morse parameter β_{eff} may appear complicated, it contains the essential coupling of the various channel states. The detailed SACM includes these effects naturally in the formulation.

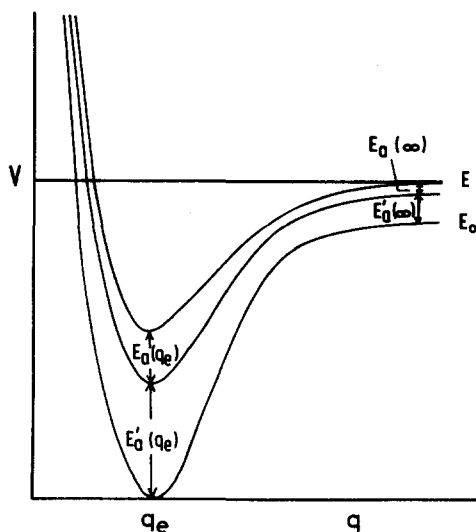


FIG. 11. Channel potential curves without centrifugal contributions for construction of D_{eff} and, hence, β_{eff} .

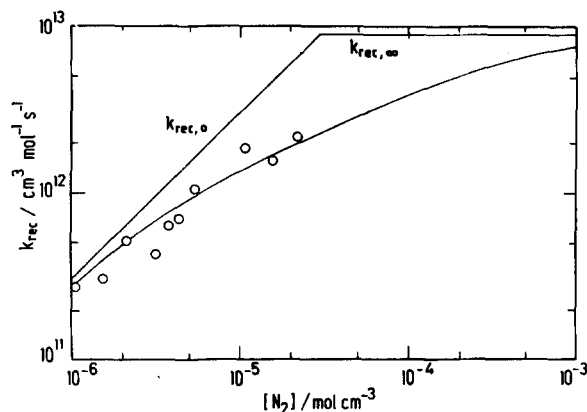


FIG. 12. Falloff curve for the thermal recombination $2 \text{ OH} \rightarrow \text{HOOH}$ at 298 K. The open circles are the measurements from Ref. 38.

APPENDIX B: FALLOFF CURVES FOR THE THERMAL DISSOCIATION-RECOMBINATION REACTION $\text{HOOH} \rightarrow 2 \text{ OH}$

We use an experimental value of the limiting high pressure rate coefficient for the recombination reaction $2 \text{ OH} \rightarrow \text{HOOH}$ at 300 K, $k_{\text{rec},\infty} = 1.5 \times 10^{11} \text{ cm}^3 \text{ molecule}^{-1} \text{ s}^{-1}$, to fit the ratio α/β in the SACM. This fitted value, $\alpha/\beta = 0.44$ using the detailed SACM and $\alpha/\beta = 0.5$ using the simplified SACM, allows us to predict the temperature dependence of both the recombination rate coefficient $k_{\text{rec},\infty}(T)$ and the dissociation rate coefficient $k_{\text{diss},\infty}(T)$. For the temperature range of 200–1500 K, we obtain

$$k_{\text{rec},\infty} = 10^{-10.82} (T/300 \text{ K})^{-0.37} \text{ cm}^3 \text{ molecule}^{-1} \text{ s}^{-1}. \quad (\text{B1})$$

Using the equilibrium constant,

$$K_c = 10^{25.71} \exp(-24\,890 \text{ K}/T) \text{ molecule cm}^{-3}, \quad (\text{B2})$$

for the range of 1000–1500 K³² yields

$$k_{\text{diss},\infty} = 10^{14.47} \exp(-24\,374 \text{ K}/T) \text{ s}^{-1} \quad (\text{B3})$$

for those temperatures.

We must construct falloff curves for the recombination and dissociation measurements since neither has been made in the high pressure limit corresponding to $k_{\text{rec},\infty}$ and $k_{\text{diss},\infty}$. We use the calculated $k(E,J)$ from the simplified SACM to obtain strong collision falloff curves that include rotational effects, a procedure that is impossible without the availability of rotational state specific rate constants. The resulting curves are well represented over the temperature range of 200–1500 K by the functional form^{43,44}

$$\frac{k}{k_\infty} = \left(\frac{k_0/k_\infty}{1 + k_0/k_\infty} \right) F_{\text{cent}}^{\{1 + [\log(k_0/k_\infty)/N]^2\}^{-1}}. \quad (\text{B4})$$

The calculated strong collision broadening factors $F_{\text{cent}}^{\text{SC}}$ for $k_0/k_\infty = 1$ are 0.63 at 200 K, 0.62 at 300 K, 0.65 at 900 K, and 0.69 at 1500 K. This parameter is nearly independent of temperature because the OH stretching vibrations make essentially no contribution to the internal energy of the activated complex $U^\ddagger$ over the specified temperature range. All of

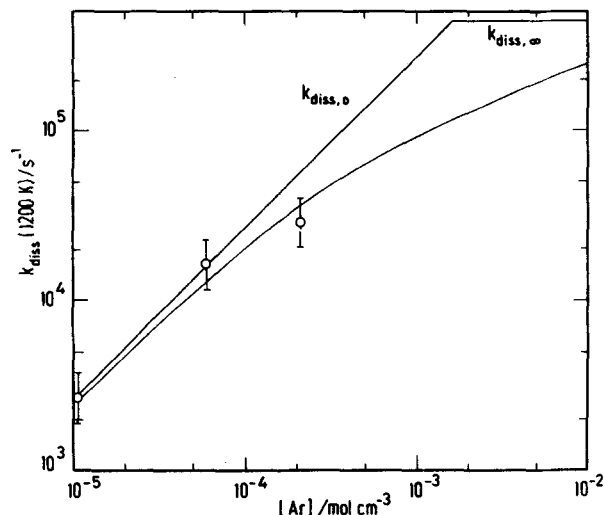


FIG. 13. Falloff curve for the thermal dissociation $\text{HOOH} \rightarrow 2 \text{ OH}$ at 1200 K. The open circles are the measurement from Ref. 39, corrected for impurities according to Ref. 45.

the product rotors are fully excited, and the overall rotation is only slightly hindered by the centrifugal barriers. Estimating the “Kassel parameter” as $S_K = 1 + U^\ddagger/kT = 4$ gives $F_{\text{cent}}^{\text{SC}} \approx 0.6$,⁴³ in good agreement with the calculations using the specific rate constants. The slight increase of $F_{\text{cent}}^{\text{SC}}$ with temperature that we obtain, for a temperature independent S_K , comes from the decrease of the B_T parameter^{43,44} as well as from some decrease of S_K because of centrifugal effects. The width factor N , which we calculate as close to unity, agrees well with the value given by the relation $N^{\text{SC}} \approx 0.75 - 1.27 \log F_{\text{cent}}^{\text{SC}}$.

There is also weak collision broadening of the falloff curves.^{43,44} For Ar as the third body, we estimate $\beta_c(300 \text{ K}) \approx 0.2$ and $\beta_c(1200 \text{ K}) \approx 0.07$ and, thus, calculate⁴⁴ that $F_{\text{cent}}^{\text{WC}}$ is about 0.8 at 300 K and 0.7 at 1200 K. The total broadening factors are roughly $F_{\text{cent}} = F_{\text{cent}}^{\text{SC}} F_{\text{cent}}^{\text{WC}} \approx 0.5 \pm 0.05$ over the range of 200–1500 K with a width of $N \approx 1.13$. Using these parameters in Eq. (B4), we reproduce the low temperature recombination and high temperature dissociation as illustrated in Figs. 12 and 13. The values of k_∞ and F_{cent} give limiting low pressure rate constants of $k_{\text{rec},0} = [\text{N}_2] 8 \times 10^{-31} (T/300 \text{ K})^{-0.76} \text{ cm}^3 \text{ molecule}^{-2} \text{ s}^{-1}$ for N_2 as a third body and $k_{\text{diss},0} = [\text{Ar}] 10^{-7.5} \exp(-21\,640 \text{ K}/T) \text{ cm}^3 \text{ molecule}^{-1} \text{ s}^{-1}$ for Ar as a third body over the range 1000–1400 K. The experimental observations of the pressure dependence are consistent with our falloff curves based only on the measured high pressure recombination rate constant at 300 K and the limiting low pressure values.

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